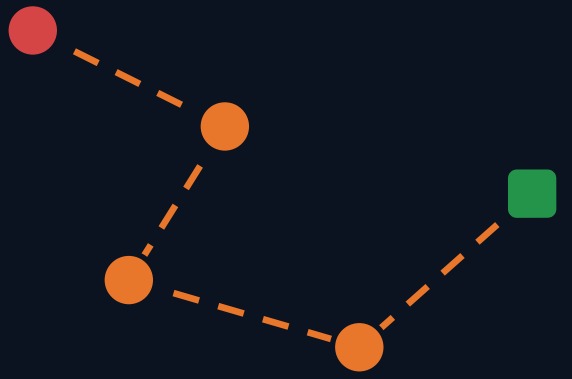
TEAM DETAILS

ChainSight

Crypto Fraud Investigation Intelligence

From a reported wallet to traceable fund flows, likely exchange links and source-backed evidence.



PROBLEM STATEMENT

Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics

Theme: Blockchain & Cybersecurity
Category: Software

TEAM EXPERTISE

Ayush Sharma

Crypto Domain & Product Research

- 1+ year of hands-on experience in cryptocurrency markets
- Understands crypto users, exchanges and market risks
- Leads problem research; turns domain knowledge into product requirements

Chhaya Attri

Full-Stack Development & Intelligence Research

- Full-stack developer: builds the dashboard, backend and APIs
- Scam-intelligence research and TinyFish research workflows
- Source verification and evidence documentation

Problem Statement

Victims report the scammer's wallet. Investigators can't quickly see which exchange (VASP) the funds may have reached, so money moves on before a request goes out.

₹22,845 cr

lost by Indians to cyber fraud in 2024, about 3× the ₹7,465 crore lost in 2023

22.68 lakh cybercrime complaints on NCRP in 2024

Source: Ministry of Home Affairs, written reply in the Lok Sabha (2025)

Fraud types

Investment · Task-based · Sextortion · Ransomware · Phishing · Darknet · Organised

Reported wallets are

Non-custodial · Temporary burners · Layering intermediaries

Manual tracing breaks on

Multi-chain transfers · DeFi protocols · Mixers / tumblers · Bridges · Privacy tools

Delay costs

Asset freezing · Evidence preservation · Fund-flow tracing · Victim recovery

Proposed Solution

Wallet in → confidence-rated, source-linked leads out



1. Ingest

Reported wallet via dashboard or API

2. Trace

Hop-by-hop transaction graph; layering patterns

3. Flag

Likely exchange-associated addresses, each confidence-rated

4. Research

Public scam reports + the exchange's official process

TinyFish

5. Report

Leads, sources and evidence for the officer to verify

TinyFish

WHAT MAKES CHAINSIGHT DIFFERENT

On-chain fund-flow analysis + TinyFish-powered discovery of public scam intelligence and official exchange law-enforcement procedures, in one source-linked investigation report. Built for investigator verification and evidence traceability: risk scores are leads, never proof.

Target Users

PRIMARY

State cyber cells and police units

Investigators handling NCRP complaints and SAHYOG requests

SECONDARY

Exchange and VASP compliance teams

Faster, standard requests about reported fraud flows

SECONDARY

Banks' fraud units

Link crypto cash-outs to the accounts they hold

VALUE TO THEM

Faster response

exchange found in one run

More freezes

request reaches the VASP sooner

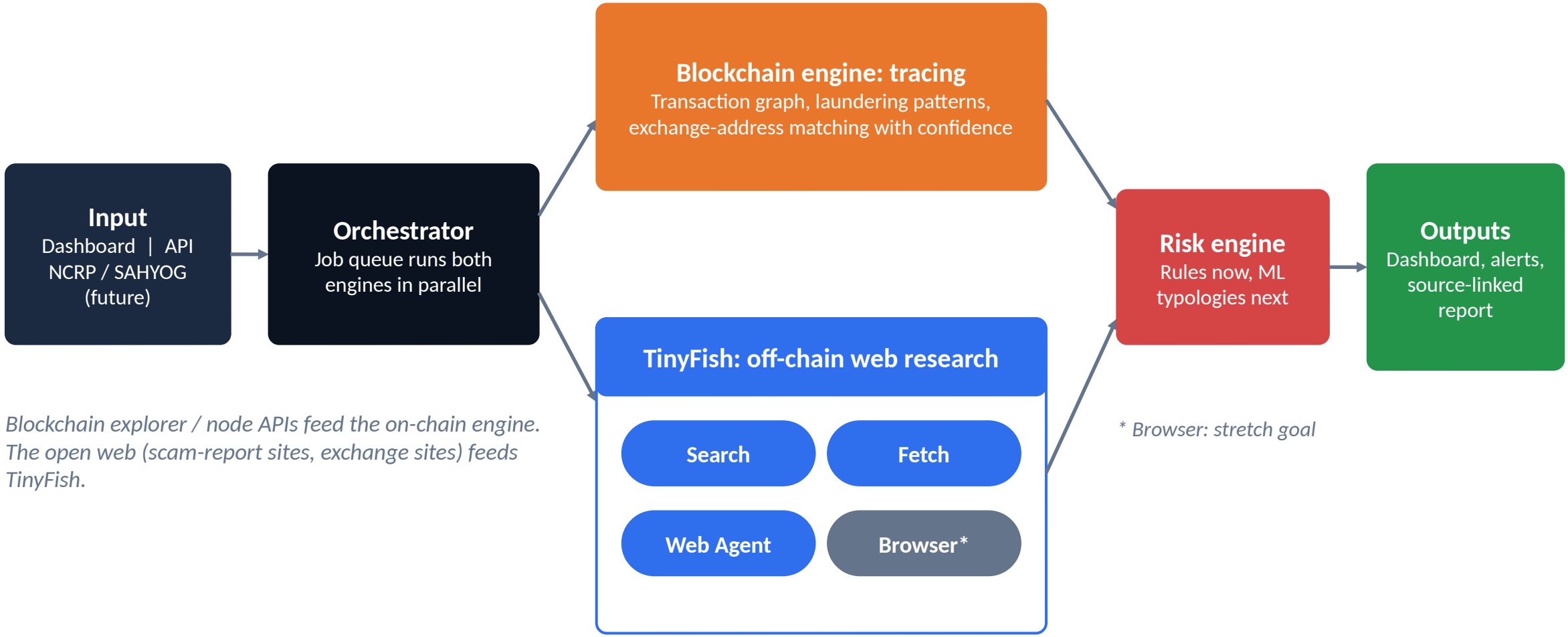
VASP coordination

right contact, standard format

Stronger evidence

timestamped, hashed snapshots

Technical Architecture and Stack



Blockchain explorer / node APIs feed the on-chain engine. The open web (scam-report sites, exchange sites) feeds TinyFish.

* *Browser: stretch goal*

STACK

Python + FastAPI | Redis worker | PostgreSQL | NetworkX graph | React dashboard | PDF / JSON reports | TinyFish API

How You Plan to Integrate the TinyFish API

The blockchain engine traces transactions. TinyFish does the off-chain web research.

	IN OUR FLOW	WHY TINYFISH
Search	Discover public scam reports about the wallet and the exchange's official pages	Structured live results with URLs
Fetch	Retrieve and extract the content of those URLs; confirm the wallet appears; log URL + time	Clean text or JSON from any page
Web Agent	Navigate the exchange's official site to its law-enforcement reporting process, where publicly accessible	Multi-step goal on any site; structured JSON; never logs in

Why not a normal scraper?

Every exchange site is different, changes often and has no API. One agent goal works across all of them; a scraper needs a script per site.

Example agent goal

"Find this exchange's law-enforcement request process. Don't log in. Return portal, steps, documents, contact." Returns JSON fields we drop into the report.

Scalability, Feasibility and Use Case

24H MVP: ONE WORKING FLOW

- One reported wallet → one report
- Tron, with live APIs where feasible
- Tested on known public wallets
- TinyFish on 1–2 exchanges
- Leads labelled with confidence

Stretch: Ethereum, deeper tracing. Later: more chains, cross-chain, ML, NCRP / SAHYOG integration (with authorised access)

Scales by configuration

- New chain or exchange = one config entry, no new scraper
- Tracing and agents run in parallel via the job queue
- VASP contacts cached and reused across cases
- REST API ready for future NCRP / SAHYOG integration, once authorised

Feasible now: existing blockchain APIs + TinyFish, no custom indexer for the MVP

Risks we plan for

Attribution is uncertain

Confidence levels; not proof the exchange holds funds

Agent fails on an odd site

Retries, then manual-review flag

Mixers and privacy tools

Flag where the trail breaks; expert review

False positives

Human review; score explained point by point

Use Case: Worked Example

A real-world precedent for coordinated tracing and freezing (not a ChainSight result)



Sources: Blockworks, OCCRP, Benzinga coverage of the Tether / OKX announcement (Nov 2023)

Where ChainSight could help an Indian cyber cell in a similar case

- Officer enters the reported wallet from the complaint
- Leads to likely exchange-linked addresses, confidence-rated
- TinyFish finds public reports and the official request process
- Officer verifies, then requests via the exchange's process

Aim: cut manual research time. We will measure it in the build, not assume it.

MVP / Prototype

ILLUSTRATIVE MOCKUP: SAMPLE DATA, NOT A REAL INVESTIGATION RESULT

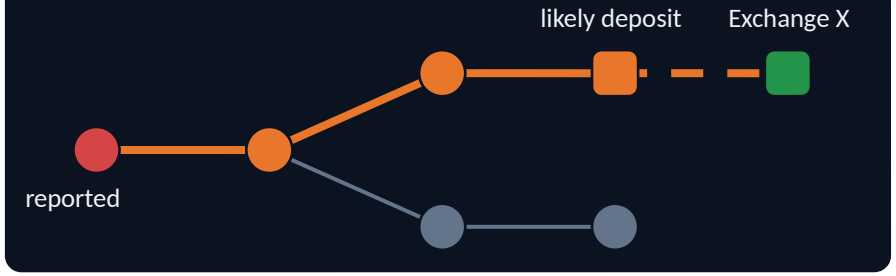
Status: design only, to be built during the hackathon. No live findings.

ChainSight Case DEMO-001 (sample) | Tron [Export report]

LEAD CONFIDENCE	LIKELY EXCHANGE-LINKED	OBSERVED TXS	SOURCES
Medium	Exchange X	3 tx hashes	4 URLs, timed

Fund flow

solid = observed tx dashed = inferred link



Exchange X: official LE process

via TinyFish Agent + source URL

Portal: LE request form
Steps: official email, notice, submit
Retrieved: 10:44 UTC

Draft request (officer verifies)

Evidence trail (sample entries)

OBSERVED	tx 9c1e…44a0 reported wallet → TQ7…9F2 8,990 USDT 08 Oct 13:27 UTC
INFERRED	TQ7…9F2 sweeps to an Exchange X wallet → likely deposit address (medium)
SOURCE	scam-report page, retrieved 10:42 UTC sha256 3f9a…c1

1. **Confidence, not verdict**
   Every lead rated High / Medium / Low
2. **Observed vs inferred**
   Tx hashes are facts; exchange links are inferences
3. **Official process**
   Found by the TinyFish agent, with its source URL
4. **Traceable evidence**
   Source URLs, retrieval times, hashes